

Data Preprocessing Techniques

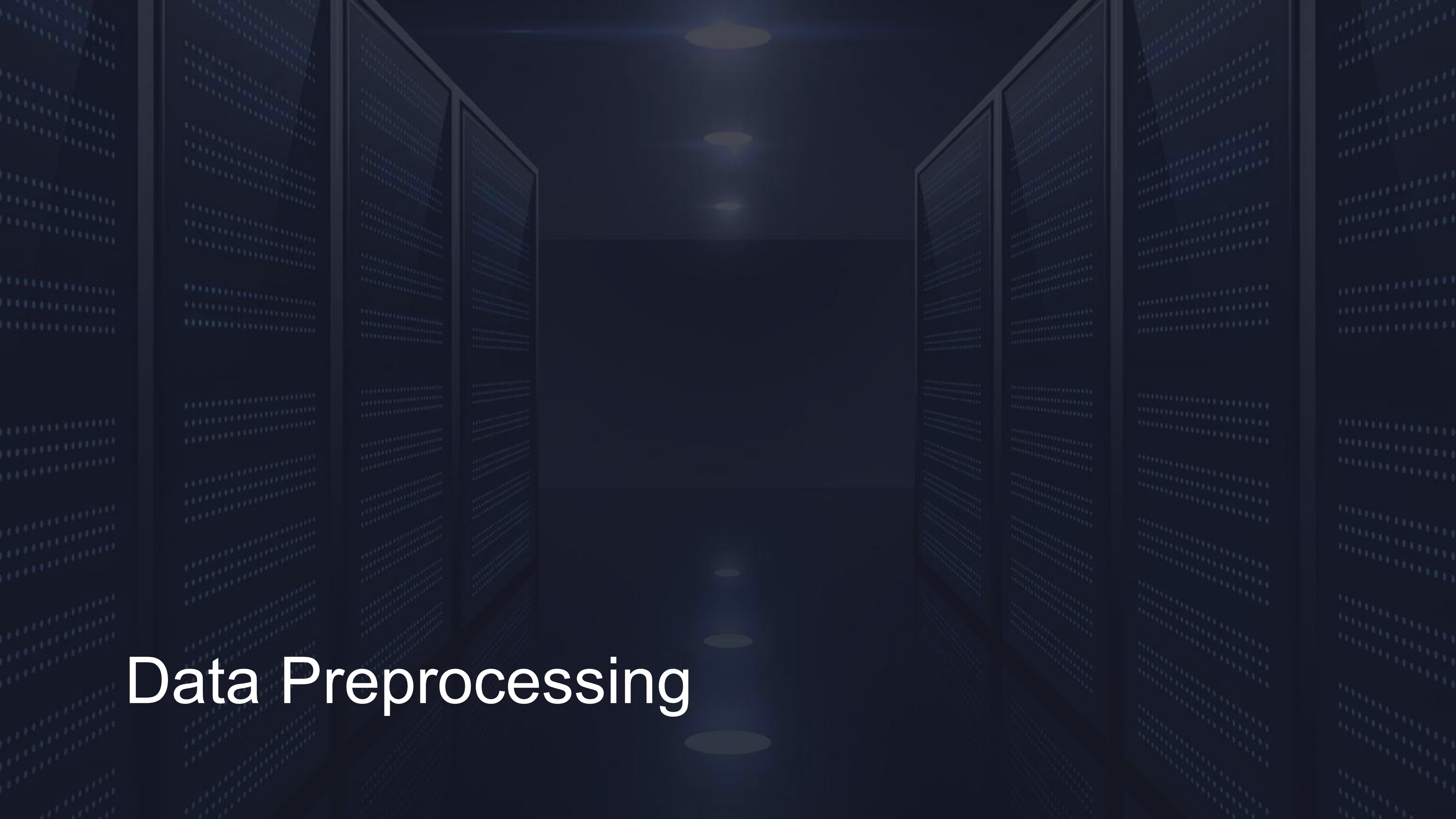
CS 415 - Data Mining

Topics to be discussed

What is data preprocessing

Data Cleaning

Data Transformation



Data Preprocessing

What is Data Preprocessing

Preprocessing is a Data Mining process that transforms raw data into useful and efficient format that the computer may use to derive information



Why preprocess data?

Data may be "Incomplete"

e.g., occupation=" "

Data may be "noisy"

e.g., Salary = "-10"

Data may be "inconsistent"

e.g., Age="42" Birthday="03/07/1997"

e.g., Was rating "1,2,3", now rating "A, B, C"

Major Preprocessing Techniques

Data cleaning

Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies

Data integration

Integration of multiple databases, data cubes, or files

Data transformation

*Normalization and aggregation and discretization

Data reduction

Obtains reduced representation in volume but produces the same or similar analytical results

Data Cleaning

Ignore the record completely

Not very effective, unless record contains several attributes with missing values

Fill in missing value manually

May not be feasible for very large data set with many missing values

Use global constant to fill in missing value

Replace missing labels with “unknown”

Use attribute mean to fill in the missing value

use average income of all consumers to replace the missing value for income

Data Noise

Binning

Sorted values are distributed into a number of buckets or bins

Regression

Using Line Regression for reducing outliers

Clustering

Groups data into multiple clusters to determine and eliminate outliers

Binning

Equal-Frequency Bins or Equal-Width

Example:

Price data : (4, 8, 15, 21, 21, 24, 25, 28, 34)

Width: 3

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Binning

Smooth by Bin Means

Example:

Price data : (4, 8, 15, 21, 21, 24, 25, 28, 34)

Width: 3

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

Binning

Smooth by Bin Boundaries

Example:

Price data : (4, 8, 15, 21, 21, 24, 25, 28, 34)

Width: 3

Bin 1: 4, 4, 15

Bin 2: 21, 21, 24

Bin 3: 25, 25, 34

Data Transformation: Min-Max Normalization

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

Where:

v' = transformed data

v = original data

$\min_A$ = Minimum value of an Attribute

$\max_A$ = Maximum value of an Attribute

new_min_A = New Minimum Value of an Attribute

new_max_A = New Maximum Value of an Attribute

Data Transformation: Min-Max Normalization

Example:

Salary Data: (12,000 , 13,487 , 13,658 , 34,784 , 42,420 , 56,487, 88,546, 98,000)

New Range: [0.0, 1.0]

New data to be transformed: 73,600

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

Data Transformation: Z-Score Normalization

$$v' = \frac{v - \mu_A}{\sigma_A}$$

Where:

v' = transformed data

v = original data

μ_A = Mean value of an Attribute

σ_A = Standard Deviation value of an Attribute

Data Transformation: Z-Score Normalization

Example:

New data to be transformed: 73,600

Mean: 54,000

Standard Deviation: 16,000

$$v' = \frac{v - \mu_A}{\sigma_A}$$

$$\frac{73,600 - 54,000}{16,000} = 1.225$$

Data Transformation: Decimal Scaling

It normalizes by moving the decimal point of values of attribute A

$$v' = \frac{v}{10^c}$$

$$\text{max_value} / 10^c < 1$$

Where:

v' = transformed data

v = original data

c = Number of decimal places of the smallest value

Data Transformation: Decimal Scaling

Example:

Range of data: [-986, 917]

$c = 3$

Data to be transformed: 547

$$v' = \frac{v}{10^c}$$

$$\frac{547}{10^3} = 0.547$$

Thank you so much!!